DATA FUNDAMENTALS PROJECT REPORT

EMPLOYEE SALARY DATA ANALYSIS

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PROBLEM STATEMENT

Companies maintain large amounts of employee data including name, department, experience, salary, and location. However, this data is often not analyzed effectively.

The objective of this project is to analyze employee salary data to identify patterns such as:

Department-wise salary distribution

Average salary in the company

Relationship between experience and salary

Highest-paid employees

This analysis helps management make better decisions regarding salary structure and workforce planning.

SOLUTION / FEATURES

This project uses SQL, Excel, and Python to analyze employee data.

Key Features:

Display and manage employee records using SQL

Calculate average salary

Identify highest-paid employees

Analyze department-wise salary

Visualize data using charts and graphs

Tools Used:

- SQL (SQlite)
- Excel (Pivot Tables, Charts)
- Python (Google Colab,Matplotlib,Pandas)

INSIGHTS

1. Highest Paying Department

Finance and Marketing departments show the highest salary ranges
Some of the top salaries (₹120K–₹130K+) belong to these departments

2. Average Salary (Approx)

Average monthly salary ≈ ₹85,000 – ₹90,000

3. Experience vs Salary

Employees with 10+ years experience generally earn higher salaries
However, a few exceptions exist (low salary despite high experience)
 Means positive correlation but not perfect

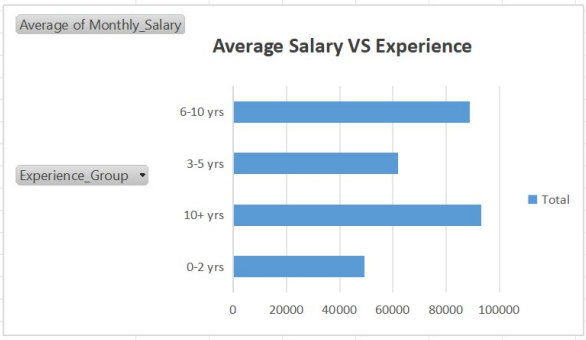
4. Highest Paid Employees

Top earners include:
Employee_38 → ₹149,123
Employee_11 → ₹141,381
Employee_25 → ₹132,455
Employee_94 → ₹132,000

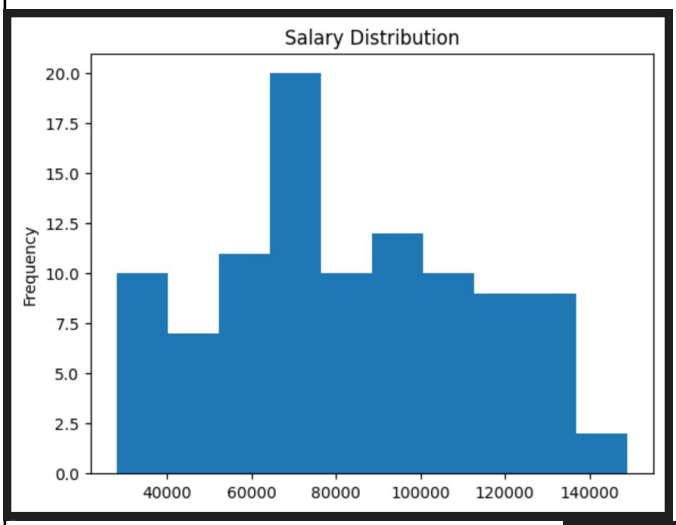
EXCEL

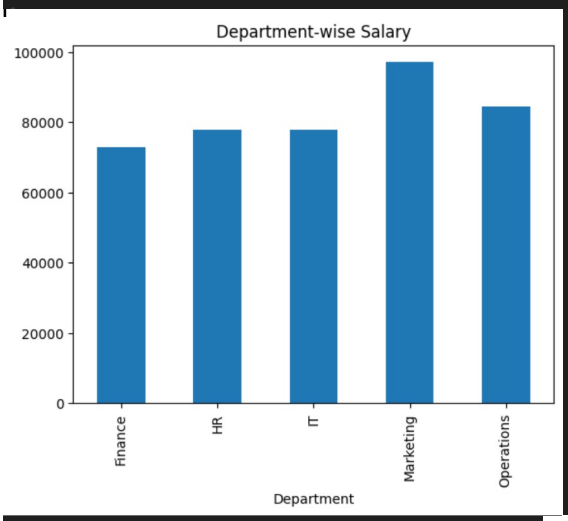
EmployeeID	Name	Department	Experience	Education_Level	Age	Gender	City	Monthly_Salary	Experience_Group
38	Employee_38	Operations	9	Master	23	Male	Mumbai	149123	6-10 yrs
11	Employee_11	Marketing	8	PhD	43	Male	Delhi	141381	6-10 yrs
34	Employee_34	HR	15	Bachelor	53	Female	Delhi	134616	10+ yrs
25	Employee_25	HR	8	High School	34	Female	Bangalore	132455	6-10 yrs
5	Employee_5	Operations	15	Master	36	Female	Delhi	132450	10+ yrs
94	Employee_94	Marketing	19	PhD	50	Female	Mumbai	132000	10+ yrs
35	Employee_35	IT	13	Master	53	Male	Bangalore	130983	10+ yrs
77	Employee_77	Finance	19	PhD	52	Male	Hyderabad	130000	10+ yrs
33	Employee_33	Marketing	9	Master	36	Male	Bangalore	127946	6-10 yrs
84	Employee_84	Marketing	18	PhD	49	Female	Mumbai	125000	10+ yrs
57	Employee_57	Finance	18	Master	50	Male	Mumbai	125000	10+ yrs
23	Employee_23	Marketing	16	High School	22	Male	Bangalore	122888	10+ yrs
50	Employee_50	Marketing	10	High School	47	Female	Bangalore	121752	6-10 yrs
99	Employee_99	Marketing	17	PhD	48	Male	Mumbai	120000	10+ yrs
61	Employee_61	IT	14	PhD	48	Male	Hyderabad	120000	10+ yrs
27	Employee_27	IT	8	Bachelor	27	Female	Delhi	119660	6-10 yrs
28	Employee_28	Finance	16	Master	29	Female	Hyderabad	118179	10+ yrs
69	Employee_69	Marketing	17	Master	46	Male	Mumbai	118000	10+ yrs
88	Employee_88	HR	16	PhD	45	Female	Chennai	115000	10+ yrs
64	Employee_64	Marketing	16	PhD	44	Female	Bangalore	115000	10+ yrs

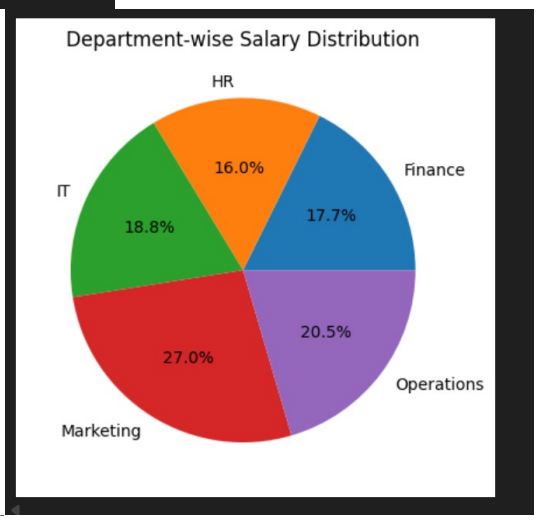
Department	Average of Monthly_Salary
Finance	73030.95
HR	77800.94118
IT	77810.25
Marketing	97069.6087
Operations	84569.95
Grand Total	82634.4

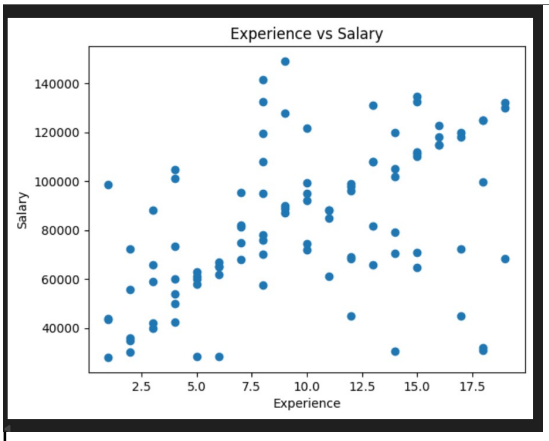


PYTHON









SQL

avg_salary
82634.4

employeeid	name	department	experience_years	education_l...	age	gender	city	monthly_salary
38	Employee_38	Operations	9	Master	23	Male	Mumbai	149123

department	total_salary
Finance	1460619
HR	1322616
IT	1556205
Marketing	2232601
Operations	1691399

UNIQUE POINTS

- Combination of SQL, Excel, and Python in one project
- Uses real-world HR dataset for analysis
- Analyzes salary based on experience and department
- Includes clear visualizations (charts & graphs)
- Converts raw data into meaningful insights

FUTURE IMPROVEMENTS

- Build dashboard using Power BI
- Apply machine learning for salary prediction
- Include performance and skill based analysis
- Use real time employee data for better accuracy